

Transformer Exercise: Neutrino Position Reconstruction

This is a short summary of my transformer model for the 2D IceCube-inspired neutrino dataset. On the test set I got an MSE loss of 2.0061, an MAE of 0.8733, and a mean position error of 1.3731. The mean position error is the average distance between the predicted and true interaction positions.

Each event is a variable-length list of photon hits. Each hit has three inputs: detection time, detector x position, and detector y position. I embedded these features, passed them through a transformer encoder, used a mask to ignore padding, mean-pooled the real hits, and predicted the interaction position (x, y) .

The transformer uses 4 attention heads and 2 encoder layers, so it satisfies the exercise requirement. The padding and masking part is needed because different events have different numbers of photon hits.

Training Curve



Figure 1: Training and validation MSE over epochs.

Overall Position Reconstruction

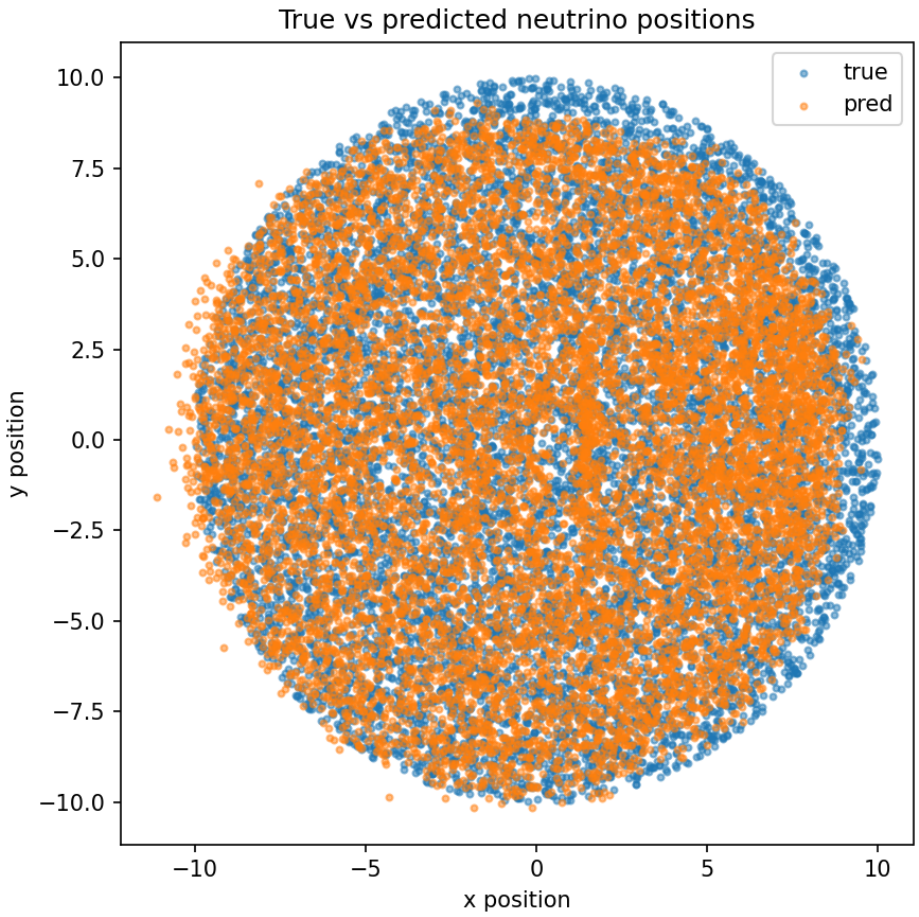


Figure 2: True and predicted neutrino positions in the (x, y) plane. The predictions cover the same general detector area as the true positions, but they are more spread out and less clean near the edges.

Position Error Distribution



Figure 3: Distribution of test position errors. Most events have a fairly small error, but there is a tail of harder events with much worse reconstruction.

Coordinate-wise Prediction Quality

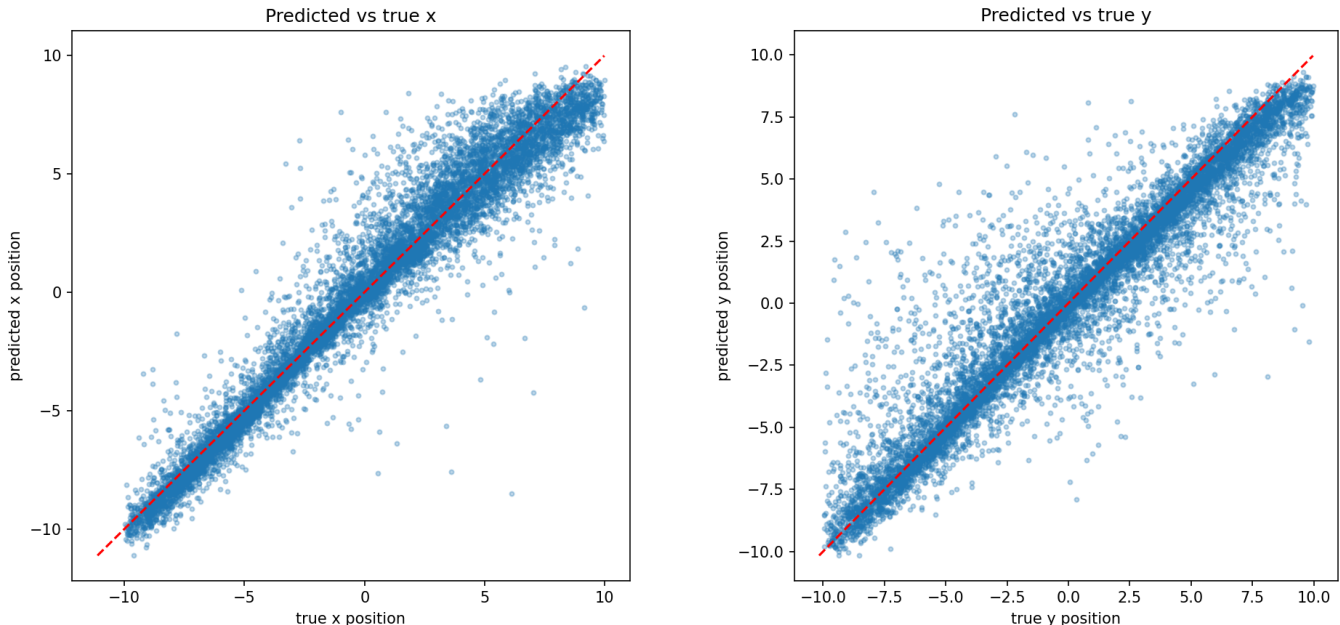


Figure 4: The dashed line is perfect prediction. The points roughly follow the line, so the model is learning the position, but the scatter shows the remaining uncertainty.

Residual Diagnostics

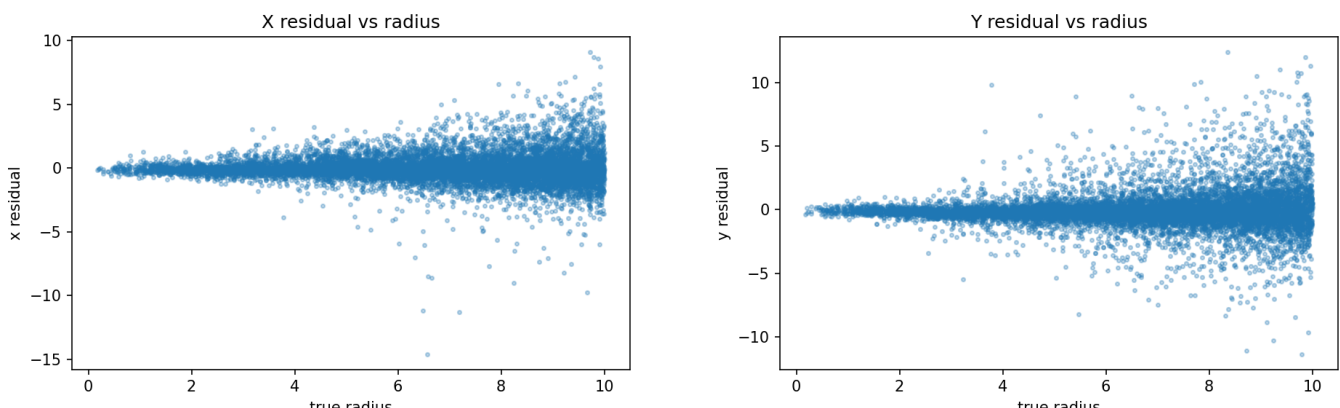


Figure 5: The residuals get wider toward the detector edge, so edge events seem harder to reconstruct. There is no obvious huge global offset.